

EE24101 Basic Electronics

Module 4: Operational Amplifiers

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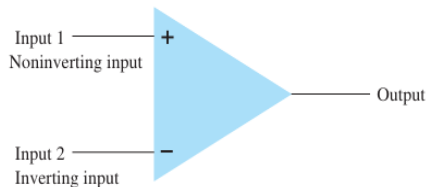
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1 Introduction

- An operational amplifier, or op-amp, is a very high gain differential amplifier with high input impedance and low output impedance.
- Typical uses of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), oscillators, filter circuits, and many types of instrumentation circuits.
- An op-amp contains a number of differential amplifier stages to achieve a very high voltage gain.
- Figure 1 shows a basic op-amp with two inputs and one output as would result using a differential amplifier input stage.
- Each input results in either the same or an opposite polarity (or phase) output, depending on whether the signal is applied to the plus (+) or the minus (-) input, respectively.



- Single-Ended Input :
- Single-ended input operation results when the input signal is connected to one input with the other input connected to ground.
- Figure 2 shows the signals connected for this operation.
- In Fig. 2(a), the input is applied to the plus input (with minus input at ground), which results in an output having the same polarity as the applied input signal.
- Figure 2(b) shows an input signal applied to the minus input, the output then being opposite in phase to the applied signal.

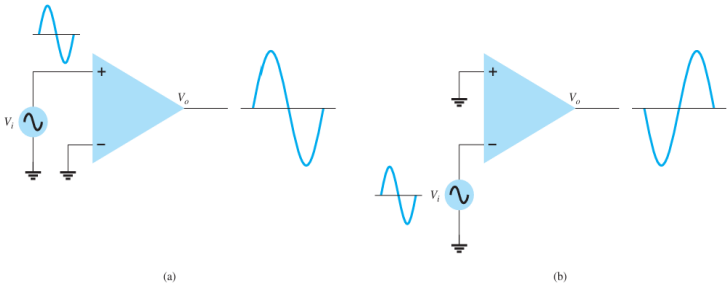
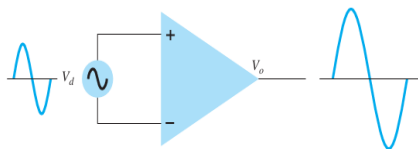
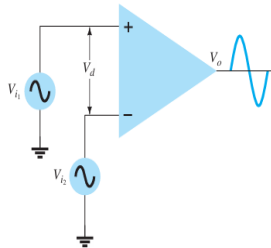


Figure 2

- Double-Ended (Differential) Input:
- In addition to using only one input, it is possible to apply signals at each input—this being a double-ended operation.
- Figure 3(a) shows an input, V_d , applied between the two input terminals (recall that neither input is at ground), with the resulting amplified output in phase with that applied between the plus and minus inputs.
- Figure 3(b) shows the same action resulting when two separate signals are applied to the inputs, the difference signal being $V_{i1} - V_{i2}$.



(a)



(b)

Figure 3

- The op-amp can also be operated with opposite outputs, as shown in Fig. 4(a).
- An input applied to either input will result in outputs from both output terminals, these outputs always being opposite in polarity.
- Figure 4(b) shows a single-ended input with a double-ended output.
- As shown, the signal applied to the plus input results in two amplified outputs of opposite polarity.

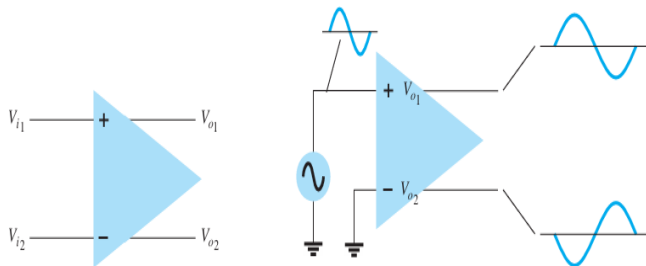
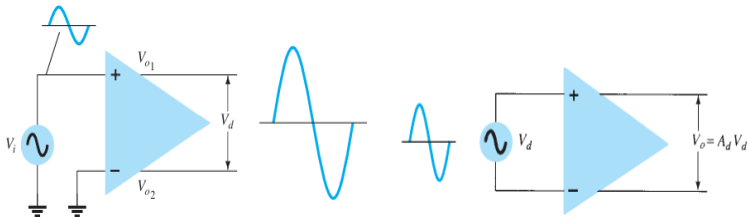


Figure 4

- Figure 5(a) shows the same operation with a single output measured between output terminals (not with respect to ground). This difference output signal is $V_{o1} - V_{o2}$.
- The difference output is also referred to as a floating signal since neither output terminal is the ground (reference) terminal.
- The difference output is twice as large as either V_{o1} or V_{o2} because they are of opposite polarity and subtracting them results in twice their amplitude [e.g., $10\text{ V} - (-10\text{ V}) = 20\text{ V}$].
- Figure 5(b) shows a differential input, differential output operation. The input is applied between the two input terminals and the output taken from between the two output terminals. This is fully differential operation.



- Common-Mode Operation:
- When the same input signals are applied to both inputs, common-mode operation results, as shown in Fig. 6.
- Ideally, the two inputs are equally amplified, and since they result in opposite-polarity signals at the output, these signals cancel, resulting in 0-V output. Practically, a small output signal will result.

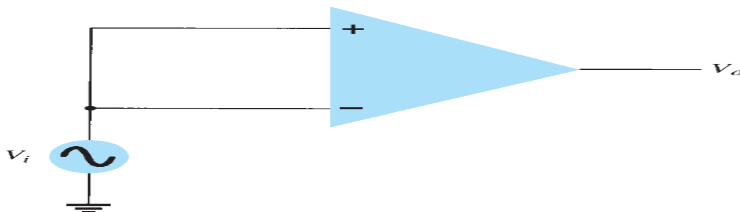
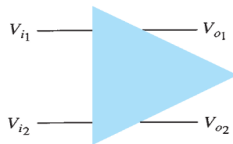
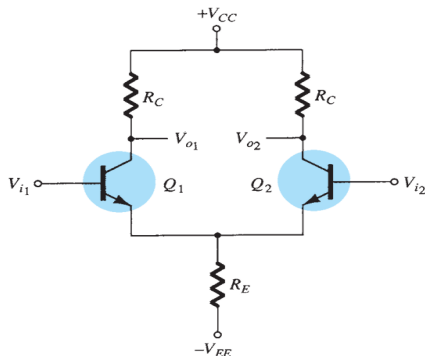


Figure 6

- A significant feature of a differential connection is that the signals that are opposite at the inputs are highly amplified, whereas those that are common to the two inputs are only slightly amplified—the overall operation being to amplify the difference signal while rejecting the common signal at the two inputs.
- Since noise (any unwanted input signal) is generally common to both inputs, the differential connection tends to provide attenuation of this unwanted input while providing an amplified output of the difference signal applied to the inputs.
- This operating feature is referred to as common-mode rejection .

Differential Amplifier Circuit

- The differential amplifier circuit shown in Fig. 7 is an extremely popular connection used in IC units.
- The circuit has two separate inputs and two separate outputs, and that the emitters are connected together.
- Whereas many differential amplifier circuits use two separate voltage supplies, the circuit can also operate using a single supply.

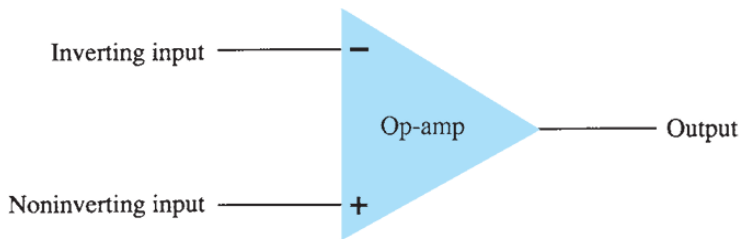


- A number of input signal combinations are possible:
- If an input signal is applied to either input with the other input connected to ground, the operation is referred to as “single-ended.”
- If two opposite-polarity input signals are applied, the operation is referred to as “double-ended.”
- If the same input is applied to both inputs, the operation is called “common-mode.”
- In single-ended operation, a single input signal is applied. However, due to the common emitter connection, the input signal operates both transistors, resulting in output from both collectors.

- In double-ended operation, two input signals are applied, the difference of the inputs resulting in outputs from both collectors due to the difference of the signals applied to both inputs.
- In common-mode operation, the common input signal results in opposite signals at each collector, these signals canceling, so that the resulting output signal is zero. The opposite signals do not completely cancel, and a small signal results.
- The main feature of the differential amplifier is the very large gain when opposite signals are applied to the inputs as compared to the very small gain resulting from common inputs.
- The ratio of this difference gain to the common gain is called common-mode rejection.

OPAMP Basics

- An operational amplifier is a very high gain amplifier having very high input impedance (typically a few megohms) and low output impedance (less than $100\ \Omega$).
- The basic circuit is made using a difference amplifier having two inputs (plus and minus) and at least one output.
- Figure 8 shows a basic op-amp unit.
- The plus (+) input produces an output that is in phase with the signal applied, whereas an input to the minus (-) input results in an opposite-polarity output.



- The ac equivalent circuit of the op-amp is shown in Fig. 9(a).
- The input signal applied between input terminals sees an input impedance R_i that is typically very high.
- The output voltage is shown to be the amplifier gain times the input signal taken through an output impedance R_o , which is typically very low.
- An ideal op-amp circuit, as shown in Fig.9 (b), would have infinite input impedance, zero output impedance, and infinite voltage gain.

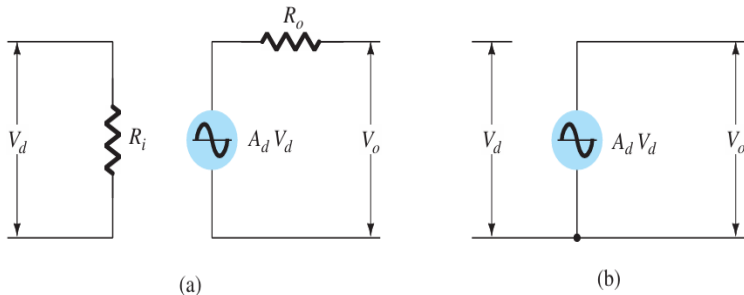


Figure 9

- The basic circuit connection using an op-amp is shown in Fig. 10.
- The circuit shown provides operation as a constant-gain multiplier.
- An input signal V_1 is applied through resistor R_1 to the minus input.
- The output is then connected back to the same minus input through resistor R_f . The plus input is connected to ground. Since the signal V_1 is essentially applied to the minus input, the resulting output is opposite in phase to the input signal.

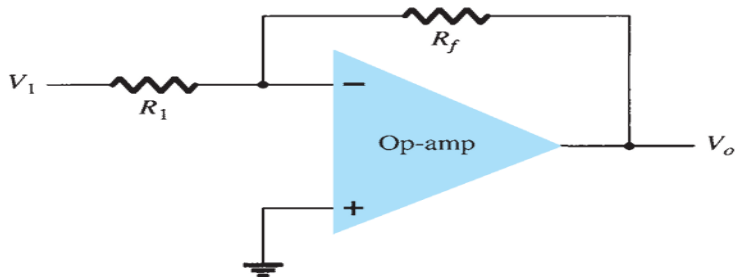


Figure 10

- Figure 11(a) shows the op-amp replaced by its ac equivalent circuit.
- If we use the ideal op-amp equivalent circuit, replacing R_i by an infinite resistance and R_o by a zero resistance, the ac equivalent circuit is that shown in Fig. 11(b).
- The circuit is then redrawn, as shown in Fig. 11(c), from which circuit analysis is carried out.

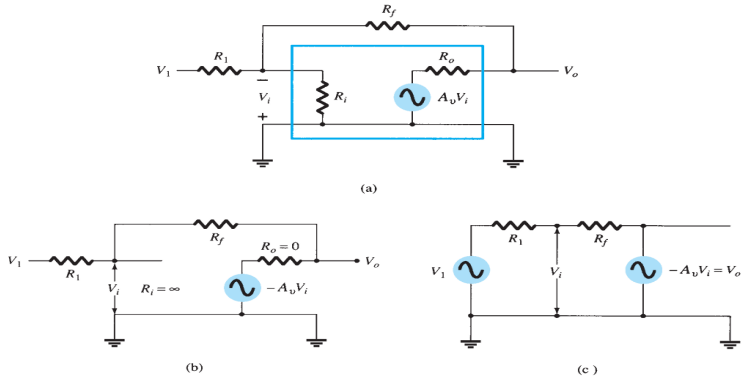


Figure 11

- Using superposition, we can solve for the voltage V_1 in terms of the components due to each of the sources. For source V_1 only ($-A_v V_i$ set to zero),

$$V_{i_1} = \frac{R_f}{R_1 + R_f} V_1 \quad (1)$$

- For source $-A_v V_i$ only (V_1 set to zero),

$$V_{i_2} = \frac{R_1}{R_1 + R_f} (-A_v V_i) \quad (2)$$

- The total voltage V_i is then

$$V_i = V_{i_1} + V_{i_2} = \frac{R_f}{R_1 + R_f} V_1 + \frac{R_1}{R_1 + R_f} (-A_v V_i) \quad (3)$$

$$V_i = \frac{R_f}{R_f + (1 + A_v)R_1} V_1 \quad (4)$$

- If $A_v \gg 1$ and $A_v R_1 \gg R_f$ then

$$V_i = \frac{R_f}{A_v R_1} V_1 \quad (5)$$

- Solving for V_o/V_i , we get

$$\frac{V_o}{V_i} = \frac{-A_v V_i}{R} V_i = \frac{-A_v R_f V_1}{V_i A_v R_1} = -\frac{R_f}{R_1} \frac{V_1}{V_i} \quad (6)$$

$$\frac{V_o}{V_i} = \frac{R_f}{R_1} \quad (7)$$

- Unity Gain: If $R_f = R_1$, the gain is

$$\text{Voltagegain} = -\frac{R_f}{R_1} = -1 \quad (8)$$

so that the circuit provides a unity voltage gain with 180° phase inversion. If R_f is exactly R_1 , the voltage gain is exactly 1.

- Constant-Magnitude Gain: If R_f is some multiple of R_1 , the overall amplifier gain is a constant. For example, if $R_f = 10R_1$, then

$$\text{Voltagegain} = -\frac{R_f}{R_1} = -10 \quad (9)$$

and the circuit provides a voltage gain of exactly 10 along with a 180° phase inversion from the input signal.

- If we select precise resistor values for R_f and R_1 , we can obtain a wide range of gains.

Virtual Ground

- The output voltage is limited by the supply voltage. As stated before, voltage gains are very high.
- If, for example, $V_o = -10\text{ V}$ and $A_v = 20,000$, the input voltage is

$$V_i = -\frac{V_o}{A_v} = \frac{10\text{ V}}{20,000} = 0.5\text{ mV} \quad (10)$$

- If the circuit has an overall gain (V_o/V_i) of, say, 1, the value of V_i is 10 V.
- Compared to all other input and output voltages, the value of V_i is then small and may be considered 0 V.
- Note that although $V_i \approx 0\text{ V}$, it is not exactly 0 V.
- The fact that $V_i \approx 0\text{ V}$ leads to the concept that at the amplifier input there exists a virtual short-circuit or virtual ground.

Virtual Ground

- The concept of a virtual short implies that although the voltage is nearly 0 V, there is no current through the amplifier input to ground.
- Figure 12 depicts the virtual ground concept. The heavy line is used to indicate that we may consider that a short exists with $V_i \approx 0V$ but that this is a virtual short so that no current goes through the short to ground.
- Current goes only through resistors R_1 and R_f as shown.

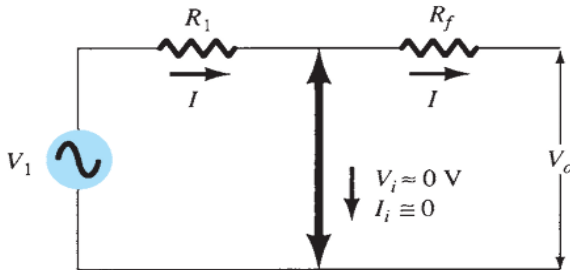


Figure 12

- Using the virtual ground concept, we can write equations for the current I as follows:

$$I = \frac{V_1}{R_1} = -\frac{V_o}{R_f} \quad (11)$$

- which can be solved for V_o/V_1 :

$$\frac{V_o}{V_1} = -\frac{R_f}{R_1} \quad (12)$$

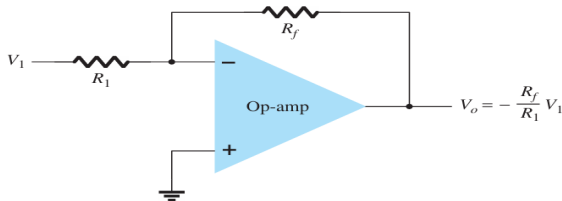
- The virtual ground concept, which depends on A_v being very large, allowed a simple solution to determine the overall voltage gain.

Practical OP-AMP Circuits

1. Inverting Amplifier:

- The most widely used constant-gain amplifier circuit is the inverting amplifier, as shown in Fig. 13.
- The output is obtained by multiplying the input by a fixed or constant gain, set by the input resistor(R_1) and feedback resistor(R_f)—this output also being inverted from the input.
- Using Eq.7, we can write

$$V_o = -\frac{R_f}{R_1} V_1 \quad (13)$$

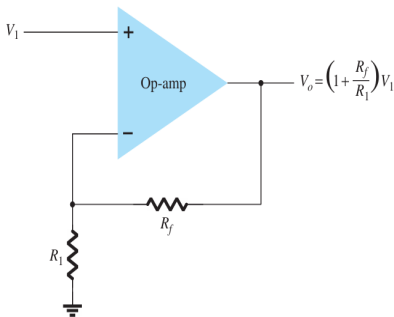


- If the circuit of Fig. 13 has $R_1 = 100 \text{ k}\Omega$ and $R_f = 500 \text{ k}\Omega$, what output voltage results for an input of $V_1 = 2 \text{ V}$?

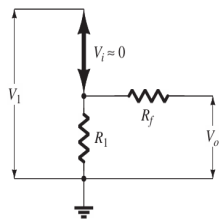
$$V_o = -\frac{R_f}{R_1} V_1 = -10 \text{ V} \quad (14)$$

• 2. Noninverting Amplifier:

- The connection of Fig. 14a shows an op-amp circuit that works as a noninverting amplifier or constant-gain multiplier.
- To determine the voltage gain of the circuit, we can use the equivalent representation shown in Fig. 14b.
- The voltage across R_1 is V_1 since $V_i \approx 0$ V.



(a)



(b)

Figure 14

- This must be equal to the output voltage, through a voltage divider of R_1 and R_f , so that

$$V_1 = \frac{R_1}{R_1 + R_f} V_o \quad (15)$$

$$\frac{V_o}{V_1} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1} \quad (16)$$

- Example: Calculate the output voltage of a noninverting amplifier (as in Fig. 14) for values of $V_1 = 2 \text{ V}$, $R_f = 500 \text{ k } \Omega$, and $R_1 = 100 \text{ k } \Omega$.
Solution:

$$V_o = \left(1 + \frac{R_f}{R_1}\right)(V_1) \quad (17)$$

$$= \left(1 + \frac{500\text{k}\Omega}{100\text{k}\Omega}\right)(2\text{V}) = +12\text{V} \quad (18)$$

Unity Follower

- The unity-follower circuit, as shown in Fig. 15a, provides a gain of unity (1) with no polarity or phase reversal.

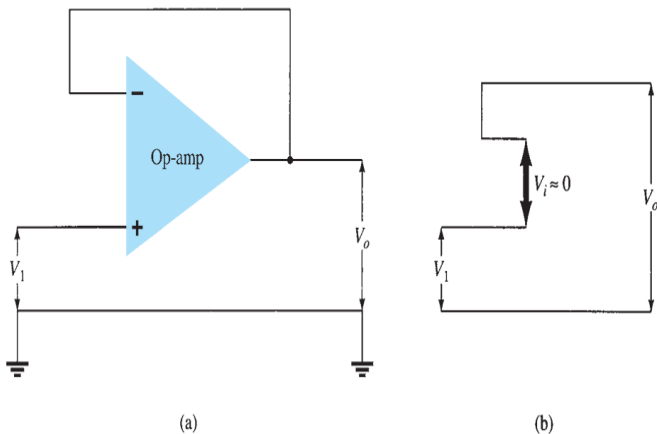


Figure 15

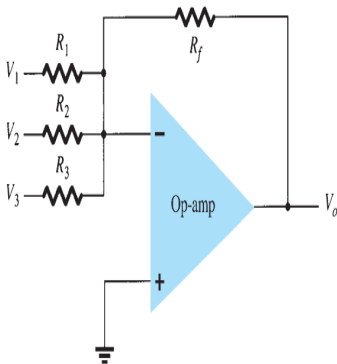
- From the equivalent circuit (see Fig. 15b) it is clear that,

$$V_o = V_i \quad (19)$$

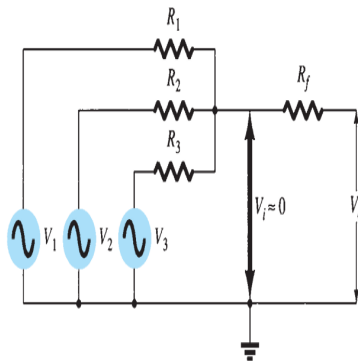
- and that the output is the same polarity and magnitude as the input.
- The circuit operates like an emitter- or source-follower circuit except that the gain is exactly unity.

Summing Amplifier

- The circuit shown in Fig. 16(a) shows a three-input summing amplifier circuit, which provides a means of algebraically summing (adding) three voltages, each multiplied by a constant-gain factor.
- Using the equivalent representation shown in Fig. 16b, we can express the output voltage in terms of the inputs as



(a)



(b)

Summing Amplifier

- Using the equivalent representation shown in Fig. 16b, we can express the output voltage in terms of the inputs as

$$V_o = -\left(\left(\frac{R_f}{R_1}\right)V_1 + \left(\frac{R_f}{R_2}\right)V_2 + \left(\frac{R_f}{R_3}\right)V_3\right) \quad (20)$$

- Each input adds a voltage to the output multiplied by its separate constant-gain multiplier.
- If more inputs are used, they each add an additional component to the output.

EXAMPLE Calculate the output voltage of an op-amp summing amplifier for the following sets of voltages and resistors. Use $R_f = 1 \text{ M}\Omega$ in all cases.

- a. $V_1 = +1 \text{ V}$, $V_2 = +2 \text{ V}$, $V_3 = +3 \text{ V}$, $R_1 = 500 \text{ k}\Omega$, $R_2 = 1 \text{ M}\Omega$, $R_3 = 1 \text{ M}\Omega$.
- b. $V_1 = -2 \text{ V}$, $V_2 = +3 \text{ V}$, $V_3 = +1 \text{ V}$, $R_1 = 200 \text{ k}\Omega$, $R_2 = 500 \text{ k}\Omega$, $R_3 = 1 \text{ M}\Omega$.

Solution: Using Eq. (10.11), we obtain

$$\begin{aligned} \text{a. } V_o &= - \left[\frac{1000 \text{ k}\Omega}{500 \text{ k}\Omega} (+1 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+2 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+3 \text{ V}) \right] \\ &= -[2(1 \text{ V}) + 1(2 \text{ V}) + 1(3 \text{ V})] = \mathbf{-7 \text{ V}} \end{aligned}$$

$$\begin{aligned} \text{b. } V_o &= - \left[\frac{1000 \text{ k}\Omega}{200 \text{ k}\Omega} (-2 \text{ V}) + \frac{1000 \text{ k}\Omega}{500 \text{ k}\Omega} (+3 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+1 \text{ V}) \right] \\ &= -[5(-2 \text{ V}) + 2(3 \text{ V}) + 1(1 \text{ V})] = \mathbf{+3 \text{ V}} \end{aligned}$$

Figure 17

Integrator

- If the feedback component used is a capacitor, as shown in Fig. 18a, the resulting connection is called an integrator.
- The virtual-ground equivalent circuit (Fig. 18b) shows that an expression for the voltage between input and output can be derived in terms of the current I from input to output.

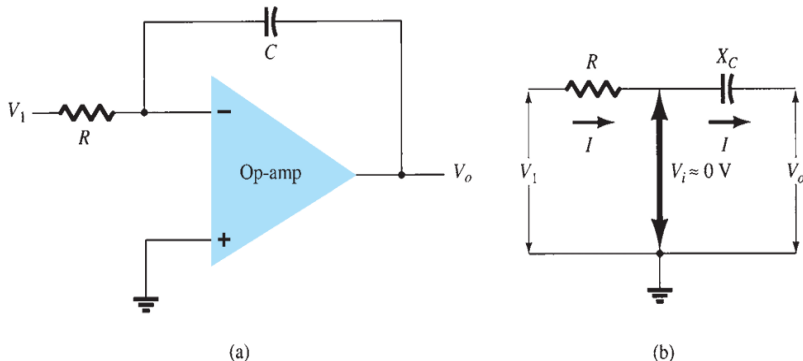


Figure 18

- Recall that virtual ground means that we can consider the voltage at the junction of R and X_C to be ground (since $V_i \approx 0$ V) but that no current goes into ground at that point.
- The capacitive impedance can be expressed as

$$X_C = \frac{1}{j\Omega C} = \frac{1}{sC} \quad (21)$$

where $s=j\Omega$ is in Laplace notation. Solving for V_o/V_1 yields

$$I = \frac{V_1}{R} = -\frac{V_o}{X_C} = -\frac{V_o}{1/sC} = -sCV_o \quad (22)$$

$$\frac{V_o}{V_i} = \frac{-1}{sCR} \quad (23)$$

- This expression can be rewritten in the time domain as

$$v_o(t) = -\frac{1}{RC} \int v_1(t) dt \quad (24)$$

Equation 25 shows that the output is the integral of the input, with an inversion and scale multiplier of $1/RC$.

Differentiator

- A differentiator circuit is shown in Fig. 19.

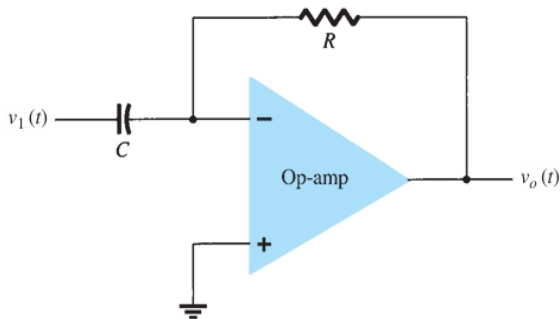


Figure 19

$$v_o(t) = -RC \frac{dv_1(t)}{dt} \quad (25)$$

OP-AMP Specifications- DC Offset Parameters

- Offset Currents and Voltages:
 - Although the op-amp output should be 0 V when the input is 0 V, in actual operation there is some offset voltage at the output.
- For example, if one connected 0 V to both op-amp inputs and then measured 26 mV(dc) at the output, this would represent 26 mV of unwanted voltage generated by the circuit and not by the input signal.
- Since the user may connect the amplifier circuit for various gain and polarity operations, however, the manufacturer specifies an input offset voltage for the op-amp.
- The output offset voltage is then determined by the input offset voltage and the gain of the amplifier, as connected by the user.
- The output offset voltage can be shown to be affected by two separate circuit conditions: (1) an input offset voltage V_{IO} and (2) an offset current due to the difference in currents resulting at the plus (+) and minus (-) inputs.

- Input Offset Voltage V_{IO} : The manufacturer's specification sheet provides a value of V_{IO} for the op-amp.
- To determine the effect of this input voltage on the output, consider the connection shown in Fig. 20.

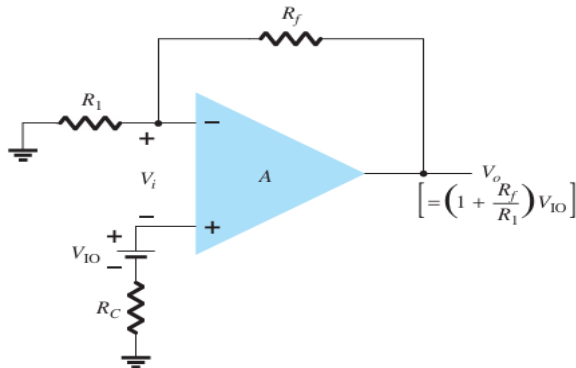


Figure 20

- Using $V_o = AV_i$, we can write

$$V_o = AV_i = A(V_{IO} - V_o \frac{R_1}{R_1 + R_2}) \quad (26)$$

Solving for V_o , we get

$$V_o = V_{IO} \frac{A}{1 + A[R_1/(R_1 + R_f)]} \approx V_{IO} \frac{A}{A[R_1/(R_1 + R_f)]}$$

from which we can write

$$V_o(\text{offset}) = V_{IO} \frac{R_1 + R_f}{R_1}$$

Figure 21

- An output offset voltage will also result due to any difference in dc bias currents at both inputs.
- Since the two input transistors are never exactly matched, each will operate at a slightly different current. For a typical op-amp connection, such as that shown in Fig. 22, an output offset voltage can be determined as follows.
- Replacing the bias currents through the input resistors by the voltage drop that each develops as shown in Fig. 23, we can determine the expression for the resulting output voltage.

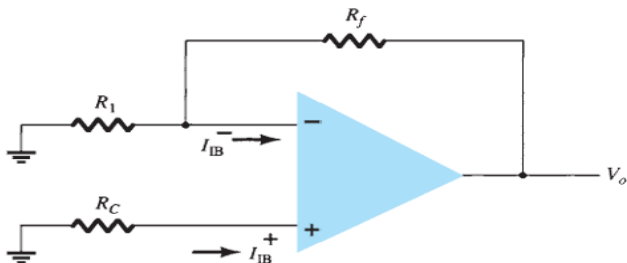


Figure 22

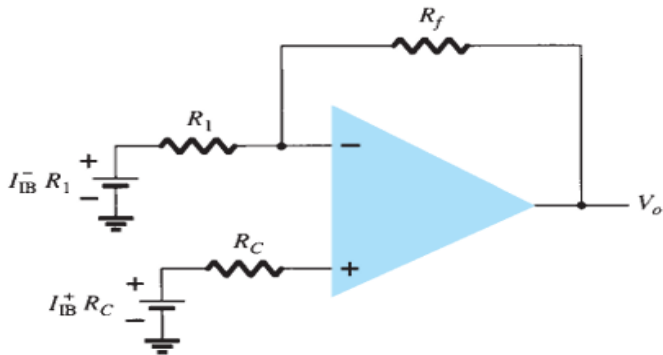


Figure 23

- Using superposition, we see that the output voltage due to input bias current I_{IB}^+ , denoted by V_o^+ , is given by

$$V_o^+ = I_{IB}^+ R_C \left(1 + \frac{R_f}{R_1}\right) \quad (27)$$

whereas the output voltage due to only I_{IB}^- , denoted by V_o^- , is given by

$$V_o^- = I_{IB}^- R_1 \left(-\frac{R_f}{R_1}\right) \quad (28)$$

for a total output offset voltage of

$$V_o(\text{offset due to } I_{IB}^+ \text{ and } I_{IB}^-) = I_{IB}^+ R_C \left(1 + \frac{R_f}{R_1}\right) - I_{IB}^- R_1 \left(\frac{R_f}{R_1}\right) \quad (29)$$

- The offset current I_{IO} is defined by

$$I_{IO} = I_{IB}^+ - I_{IB}^- \quad (30)$$

- Since the compensating resistance R_C is usually approximately equal to the value of R_1 , we can write

$$V_o(\text{offset}) = I_{IB}^+(R_1 + R_f) - I_{IB}^- R_f = I_{IB}^+ R_f - I_{IB}^- R_f = (I_{IB}^+ - I_{IB}^-) R_f \quad (31)$$

resulting in

$$V_o(\text{offset due to } I_{IO}) = I_{IO} R_f \quad (32)$$

- Input Bias Current (I_{IB}): A parameter related to I_{IO} and the separate input bias currents I_{IB}^+ and I_{IB}^- is the average bias current defined as

$$I_{IB} = \frac{I_{IB}^+ + I_{IB}^-}{2} \quad (33)$$

- Slew Rate: Another parameter reflecting the op-amp's ability to handle varying signals is the slew rate, defined as
Slew rate = maximum rate at which amplifier output can change in volts per microsecond ($V/\mu s$)

$$SR = \frac{\Delta V_o}{\Delta t} V/\mu s \quad (34)$$

- The slew rate provides a parameter specifying the maximum rate of change of the output voltage when driven by a large step-input signal.

EXAMPLE For an op-amp having a slew rate of $SR = 2 \text{ V}/\mu\text{s}$, what is the maximum closed-loop voltage gain that can be used when the input signal varies by 0.5 V in $10 \mu\text{s}$?

Solution: Since $V_o = A_{CL}V_i$, we can use

$$\frac{\Delta V_o}{\Delta t} = A_{CL} \frac{\Delta V_i}{\Delta t}$$

from which we get

$$A_{CL} = \frac{\Delta V_o / \Delta t}{\Delta V_i / \Delta t} = \frac{SR}{\Delta V_i / \Delta t} = \frac{2 \text{ V}/\mu\text{s}}{0.5 \text{ V}/10 \mu\text{s}} = 40$$

Any closed-loop voltage gain of magnitude greater than 40 would drive the output at a rate greater than the slew rate allows, so the maximum closed-loop gain is 40.

Figure 24

Differential and Common Mode Operation

- One of the more important features of a differential circuit connection, as provided in an op-amp, is the circuit's ability to greatly amplify signals that are opposite at the two inputs while only slightly amplifying signals that are common to both inputs.
- An op-amp provides an output component that is due to the amplification of the difference of the signals applied to the plus and minus inputs and a component due to the signals common to both inputs.
- Since amplification of the opposite input signals is much greater than that of the common input signals, the circuit provides a common-mode rejection as described by a numerical value called the common-mode rejection ratio (CMRR).

- **Differential Inputs:** When separate inputs are applied to the op-amp, the resulting difference signal is the difference between the two inputs.

$$V_d = V_{i_1} - V_{i_2} \quad (35)$$

- **Common Inputs:** When both input signals are the same, a common signal element due to the two inputs can be defined as the average of the sum of the two signals.

$$V_c = \frac{1}{2}(V_{i_1} + V_{i_2}) \quad (36)$$

- **Output Voltage:** Since any signals applied to an op-amp in general have both in-phase and out-of-phase components, the resulting output can be expressed as

$$V_o = A_d V_d + A_c V_c \quad (37)$$

where V_d = difference voltage; V_c = common voltage; A_d = differential gain of the amplifier; A_c = common-mode gain of the amplifier

Common-Mode Rejection Ratio (CMRR)

- $CMRR = \frac{A_d}{A_c}$
- The value of CMRR can also be expressed in logarithmic terms as

$$CMRR(\log) = 20 \log_{10} \frac{A_d}{A_c}$$

EXAMPLE

Calculate the CMRR for the circuit measurements shown in Fig. 10.52.

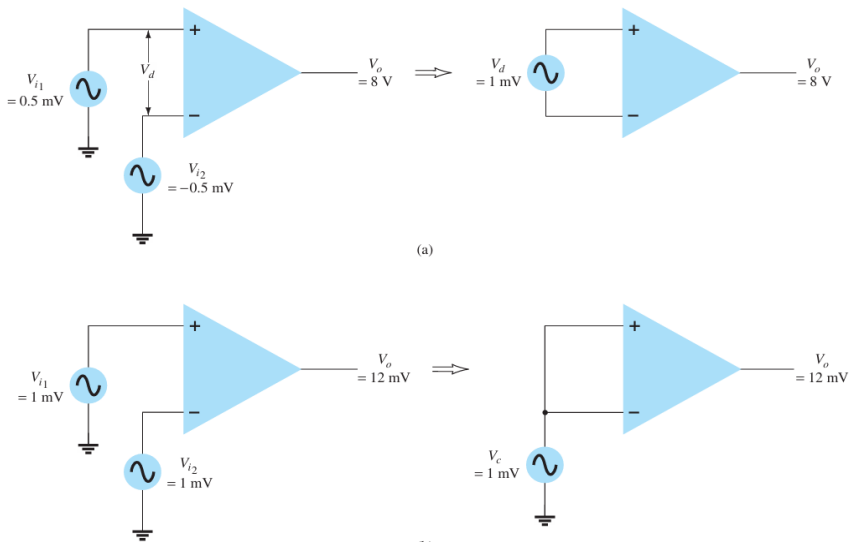


Figure 25

- Solution:
- $A_d = V_o/V_d = 8 \text{ V}/1 \text{ mV} = 8000.$
- $A_c = V_o/V_c = 12 \text{ mV}/1 \text{ mV} = 12$
- $\text{CMRR} = A_d/A_c = 8000/12 = 666.7$
- which can also be expressed as
 $\text{CMRR} = 20 \log_{10} A_d/A_c = 20 \log_{10} 666.7 = 56.48 \text{ dB}$

- The desired operation will have A_d very large with A_c very small.
- That is, the signal components of opposite polarity will appear greatly amplified at the output, whereas the signal components that are in phase will mostly cancel out so that the common-mode gain A_c is very small.
- Ideally, the value of the CMRR is infinite.
- Practically, the larger the value of CMRR, the better is the circuit operation.

- We can express the output voltage in terms of the value of CMRR as follows:

$$V_o = A_d V_d + A_c V_c = A_d V_d \left(1 + \frac{A_c V_c}{A_d V_d}\right) \quad (38)$$

$$V_o = A_d V_d \left(1 + \frac{1}{CMRR} \frac{V_c}{V_d}\right) \quad (39)$$

EXAMPLE Determine the output voltage of an op-amp for input voltages of $V_{i_1} = 150 \mu\text{V}$ and $V_{i_2} = 140 \mu\text{V}$. The amplifier has a differential gain of $A_d = 4000$ and the value of CMRR is:

- 100.
- 10^5 .

Solution:

$$\text{Eq. (10.26): } V_d = V_{i_1} - V_{i_2} = (150 - 140) \mu\text{V} = 10 \mu\text{V}$$

$$\text{Eq. (10.27): } V_c = \frac{1}{2}(V_{i_1} + V_{i_2}) = \frac{150 \mu\text{V} + 140 \mu\text{V}}{2} = 145 \mu\text{V}$$

a.
$$\begin{aligned} \text{Eq. (10.31): } V_o &= A_d V_d \left(1 + \frac{1}{\text{CMRR}} \frac{V_c}{V_d} \right) \\ &= (4000)(10 \mu\text{V}) \left(1 + \frac{1}{100} \frac{145 \mu\text{V}}{10 \mu\text{V}} \right) \\ &= 40 \text{ mV}(1.145) = \mathbf{45.8 \text{ mV}} \end{aligned}$$

b.
$$V_o = (4000)(10 \mu\text{V}) \left(1 + \frac{1}{10^5} \frac{145 \mu\text{V}}{10 \mu\text{V}} \right) = 40 \text{ mV}(1.000145) = \mathbf{40.006 \text{ mV}}$$

Figure 26

Key characteristics of an ideal op-amp

- Infinite open-loop gain: The output voltage is amplified infinitely with any tiny difference between the input voltages.
- Infinite input impedance: This prevents any current from flowing into the input terminals, meaning the op-amp does not "load" the signal source.
- Zero output impedance: The op-amp can drive any load without its output voltage dropping.
- Infinite bandwidth: The op-amp can amplify signals of any frequency with equal gain.
- Zero output offset voltage: When the input voltage difference is zero (i.e., both inputs are equal), the output voltage is exactly zero.
- Infinite slew rate: The output voltage can change instantaneously in response to a change in the input voltage.
- Infinite CMRR: This means the op-amp has an infinite ability to reject common-mode noise while amplifying the differential input signal.
- Zero input bias current: No current flows into the input terminals.

- Q1. What is the significance of virtual ground in an op-amp? Also for an ideal op-amp having a slew rate of $2\text{V}/\mu\text{s}$, determine the maximum closed-loop voltage gain that can be used when the input signal varies by 0.5 V in $10\ \mu\text{s}$?
- Q2.

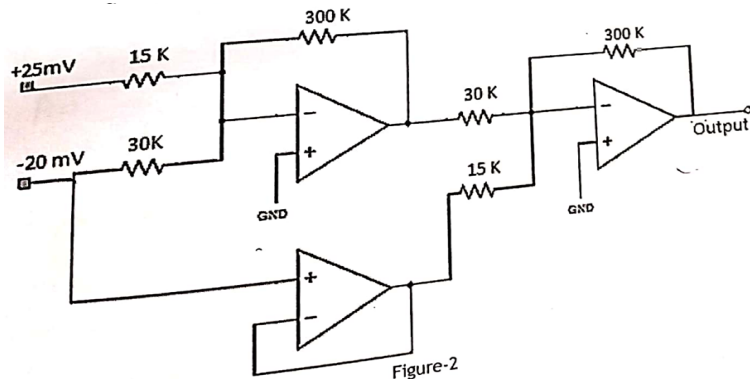


Figure 27

END OF MODULE 4